

Friction

INTRODUCTION TO FRICTION

- Friction is a force that opposes relative motion between two surfaces
 - There are several types of friction, including static, kinetic, and rolling friction
 - The force of friction depends on the nature of the surfaces in contact
 - The force of friction depends on the force pressing the surfaces together
 - The force of friction can be affected by the velocity of the moving surface
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LAWS OF FRICTION

- The force of friction is proportional to the force pressing the surfaces together
 - The force of friction is independent of the area of contact
 - The force of friction is proportional to the coefficient of friction
 - The coefficient of friction is a constant that depends on the nature of the surfaces in contact
 - The coefficient of friction can be measured experimentally
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TYPES OF FRICTION

- Static friction occurs when an object is stationary
 - Kinetic friction occurs when an object is moving
 - Rolling friction occurs when an object is rolling
 - The force of friction is proportional to the force pressing the surfaces together
 - The force of friction is independent of the area of contact
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FACTORS AFFECTING FRICTION

- The roughness of the surfaces in contact can affect the force of friction
 - The properties of the materials in contact can affect the force of friction
 - The normal force can affect the force of friction
 - The velocity of the moving surface can affect the force of friction
 - The force of friction can decrease at high velocities due to the formation of a fluid film
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APPLICATIONS OF FRICTION

- Friction is used in braking systems to slow down or stop vehicles
 - Friction is used in tires to provide traction between the tires and the road surface
 - Friction is used in clutches to transmit power from the engine to the transmission
 - The force of friction is used to convert kinetic energy into heat energy
 - The force of friction is used to accelerate, brake, and corner vehicles
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